

CHAPTER 3 – GEOGRAPHIC INFORMATION SYSTEM (GIS) AND QGIS SOFTWARE

3.1 Introduction

Geographic Information Systems (GIS) have become an essential tool for the acquisition, management, analysis, and visualization of spatial data. The integration of GIS with remote sensing technology has significantly enhanced the capability of researchers and decision-makers to understand, monitor, and manage natural and human-induced phenomena. GIS provides a framework for collecting, storing, processing, analyzing, and displaying geographically referenced information in a systematic manner.

In remote sensing applications, GIS is widely used for image interpretation, land-use and land-cover mapping, environmental monitoring, resource management, urban planning, and disaster assessment. The ability of GIS to integrate satellite imagery with other spatial datasets makes it an indispensable technology in geospatial studies.

In the present study, GIS is utilized for managing Landsat 9 satellite imagery, generating RGB composites, creating study area subsets, and preparing datasets for vegetation analysis and image classification. QGIS, an open-source GIS platform, is employed as the primary software for performing these operations.

3.2 Geographic Information System (GIS)

3.2.1 Definition of GIS

A Geographic Information System (GIS) is a computer-based system designed to capture, store, manage, analyze, and visualize spatial and geographic data. GIS combines hardware, software, data, people, and analytical methods to support spatial decision-making and problem-solving.

According to the Environmental Systems Research Institute (ESRI), GIS is a system that allows users to create interactive queries, analyze spatial information, edit data, create maps, and present the results of geographic operations.

GIS enables the integration of spatial and attribute information, thereby facilitating comprehensive analysis of geographic phenomena. It provides an efficient means of understanding spatial relationships, patterns, and trends that may not be apparent through conventional data analysis techniques.

3.2.2 Components of GIS

A GIS consists of five fundamental components that work together to perform spatial analysis and data management.

3.2.2.1 Hardware

Hardware refers to the physical infrastructure required for operating a GIS. It includes computers, servers, storage devices, GPS receivers, scanners, plotters, and networking equipment. Modern GIS applications require sufficient processing power and storage capacity to handle large geospatial datasets and perform complex analyses.

3.2.2.2 Software

GIS software provides tools for data input, storage, manipulation, analysis, and visualization. Examples of GIS software include QGIS, ArcGIS, GRASS GIS, and MapInfo. In this project, QGIS is used due to its open-source nature, extensive functionality, and compatibility with remote sensing data.

3.2.2.3 Data

Data form the foundation of any GIS. GIS datasets may include satellite imagery, topographic maps, vector layers, digital elevation models (DEMs), and attribute databases. Spatial data are generally classified into raster and vector formats.

3.2.2.4 People

GIS technology requires skilled users to collect data, perform analysis, interpret results, and make decisions. GIS professionals, researchers, planners, analysts, and policymakers play important roles in utilizing GIS for various applications.

3.2.2.5 Methods

Methods refer to the procedures, workflows, and analytical techniques used in GIS. Standardized methodologies ensure consistency, accuracy, and reliability in spatial analysis and decision-making processes.

3.2.3 Functions of GIS

GIS performs a wide range of functions that facilitate the management and analysis of spatial information.

3.2.3.1 Data Acquisition

GIS enables the collection and import of spatial data from multiple sources, including satellite imagery, GPS observations, aerial photographs, and field surveys.

3.2.3.2 Data Storage and Management

GIS provides efficient mechanisms for storing, organizing, updating, and retrieving geospatial information. Large datasets can be managed systematically within GIS databases.

3.2.3.3 Spatial Analysis

Spatial analysis is one of the most important functions of GIS. It involves the examination of spatial relationships, patterns, and trends. Common GIS analyses include overlay analysis, buffering, proximity analysis, and terrain analysis.

3.2.3.4 Visualization and Mapping

GIS enables the creation of maps, charts, and graphical representations that facilitate the interpretation and communication of spatial information.

3.2.4 Applications of GIS

GIS is widely used across various disciplines for planning, management, monitoring, and decision-making.

3.2.4.1 Urban Planning

GIS supports urban planning activities such as land-use zoning, infrastructure development, transportation planning, and population analysis.

3.2.4.2 Agriculture

GIS is extensively used for crop monitoring, precision agriculture, irrigation management, soil analysis, and yield estimation.

3.2.4.3 Environmental Monitoring

Environmental applications of GIS include forest monitoring, water resource management, pollution assessment, biodiversity conservation, and climate studies.

3.2.4.4 Disaster Management

GIS assists in disaster preparedness, risk assessment, emergency response planning, flood mapping, earthquake analysis, and damage assessment.

3.2.4.5 Mining and Geological Studies

GIS is widely employed in geological mapping, mineral exploration, mining management, and terrain analysis. The technology enables the integration of geological, topographic, and satellite datasets for effective resource assessment and planning.

3.3 QGIS Software

3.3.1 Introduction to QGIS

QGIS (Quantum Geographic Information System) is a free and open-source Geographic Information System software used for viewing, editing, analyzing, and managing geospatial data. It provides comprehensive tools for handling both raster and vector datasets and is widely used in academic research, government organizations, and industry applications.

QGIS supports a wide range of geospatial data formats and offers numerous built-in tools for map creation, spatial analysis, geoprocessing, remote sensing, and cartographic visualization. Due to its user-friendly interface and extensive functionality, QGIS has become one of the most widely adopted GIS platforms worldwide.

In the present study, QGIS was utilized for loading Landsat 9 imagery, generating RGB composite images, creating study area subsets, and preparing datasets for subsequent image analysis and classification.

3.3.2 History and Development of QGIS

QGIS was initially developed by Gary Sherman in 2002 as an open-source alternative to proprietary GIS software. Over the years, it has evolved into a powerful GIS platform supported by a global community of developers, researchers, and organizations.

The software is developed under the Open Source Geospatial Foundation (OSGeo) and is continuously updated with new functionalities and improvements. Today, QGIS is available for Windows, Linux, macOS, and several other operating systems.

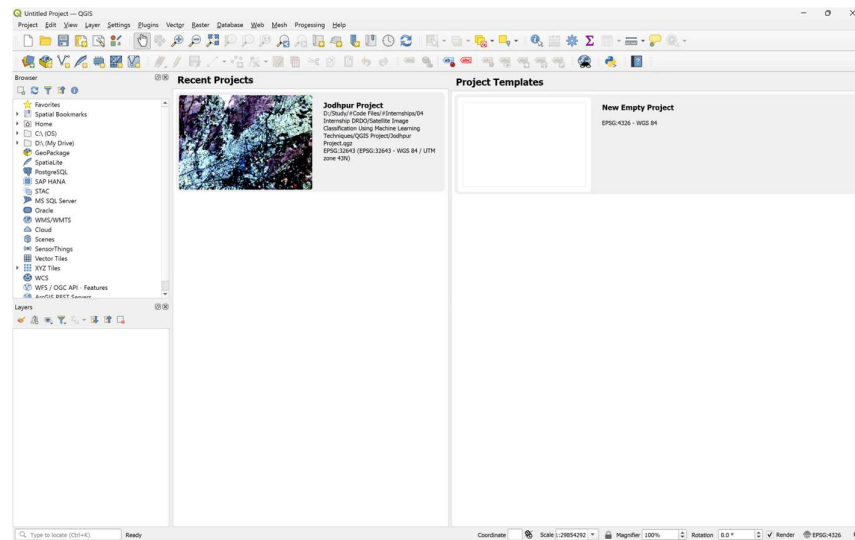


Figure 3.1: QGIS Software Environment.

3.3.3 Features of QGIS

QGIS provides numerous features that make it suitable for geospatial data analysis and remote sensing applications. Some important features include:

- Support for raster and vector data formats.
- Advanced spatial analysis tools.
- Georeferencing and coordinate system management.
- Map creation and cartographic design.
- Integration with GPS and remote sensing datasets.
- Plugin-based architecture for extending functionality.
- Compatibility with various databases and web services.
- Python scripting support for automation and customization.

3.3.4 Advantages of QGIS

The major advantages of QGIS are:

- Open-source and freely available.
- User-friendly graphical interface.
- Cross-platform compatibility.
- Extensive community support.
- Large collection of plugins and extensions.
- Capability to handle large geospatial datasets.
- Integration with remote sensing and machine learning workflows.

3.3.5 Applications of QGIS in Remote Sensing

QGIS plays a significant role in remote sensing studies by providing tools for image visualization, preprocessing, spatial analysis, and map production. Common remote sensing applications of QGIS include:

- RGB composite generation.
- Satellite image visualization.
- Land-use and land-cover mapping.
- Vegetation analysis using NDVI.
- Change detection studies.
- Image classification and thematic mapping.
- Environmental monitoring and resource management.

3.4 Installation and Configuration of QGIS

3.4.1 System Requirements

Before installing QGIS, it is necessary to ensure that the computer system meets the minimum hardware and software requirements. QGIS can operate on modern computers running Windows, Linux, or macOS operating systems.

Typical requirements include:

- 64-bit operating system.
- Minimum 4 GB RAM.
- At least 2 GB free disk space.
- Stable internet connection for downloading software and plugins.
- Multi-core processor for efficient geospatial processing.

3.4.2 Downloading QGIS

The latest version of QGIS can be downloaded from the official QGIS website (<https://qgis.org>). The website provides installers for various operating systems and also offers Long-Term Release (LTR) versions for stable production environments.

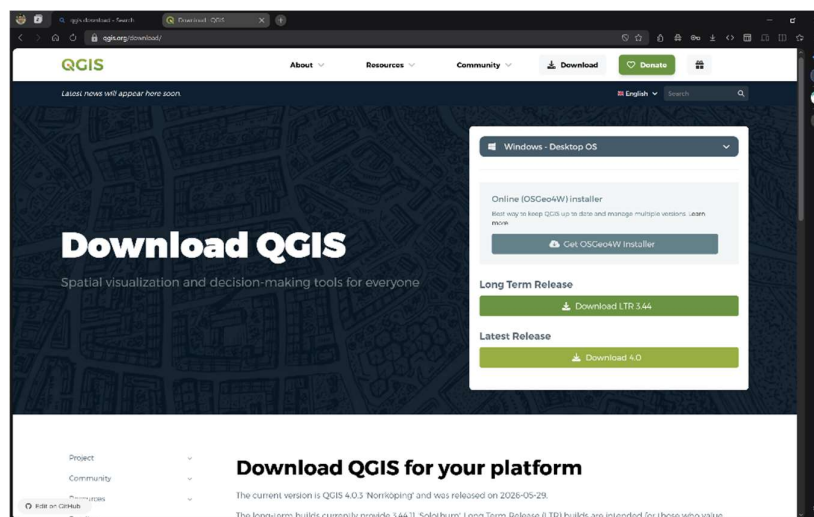


Figure 3.2: Official QGIS Download Page

The QGIS installer was downloaded from the official website and stored on the local system for installation

3.4.3 Installation Procedure

The installation of QGIS was performed using the downloaded installer package. The following steps were followed:

1. Download the QGIS installer from the official website.
2. Execute the installer package.
3. Accept the license agreement.
4. Select the installation location.
5. Choose the required components.
6. Complete the installation process.
7. Launch QGIS after successful installation.

The installation process automatically configured the necessary libraries and dependencies required for GIS operations

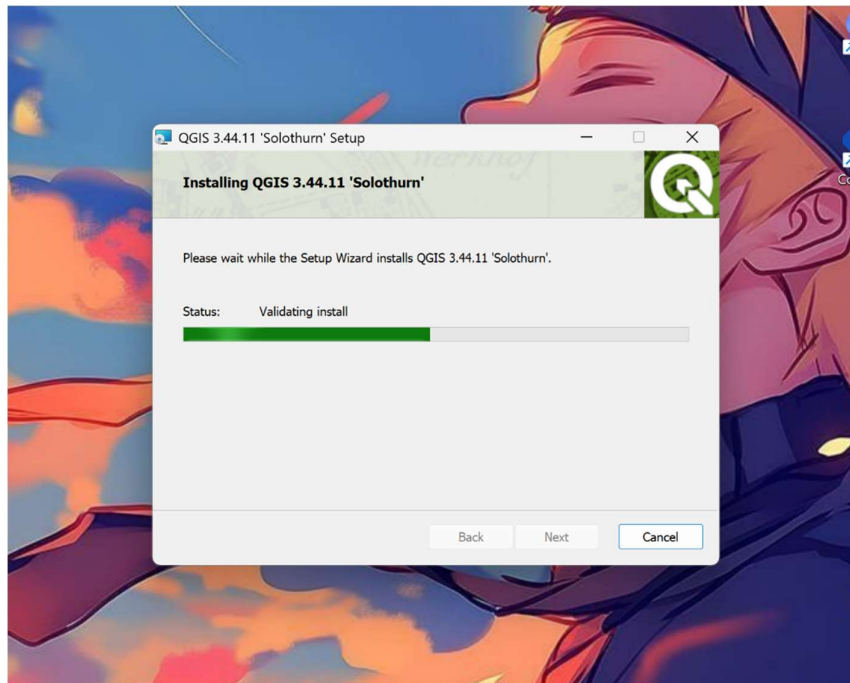


Figure 3.3: QGIS Installation Wizard

3.4.4 Initial Configuration of QGIS

After installation, QGIS was launched and configured for the project. The following settings were verified:

- Coordinate Reference System (CRS).
- Default project settings.
- Processing toolbox availability.
- Browser panel accessibility.
- Raster and vector data support.
- Plugin management options.

Proper configuration ensures efficient handling of satellite imagery and geospatial datasets throughout the project.

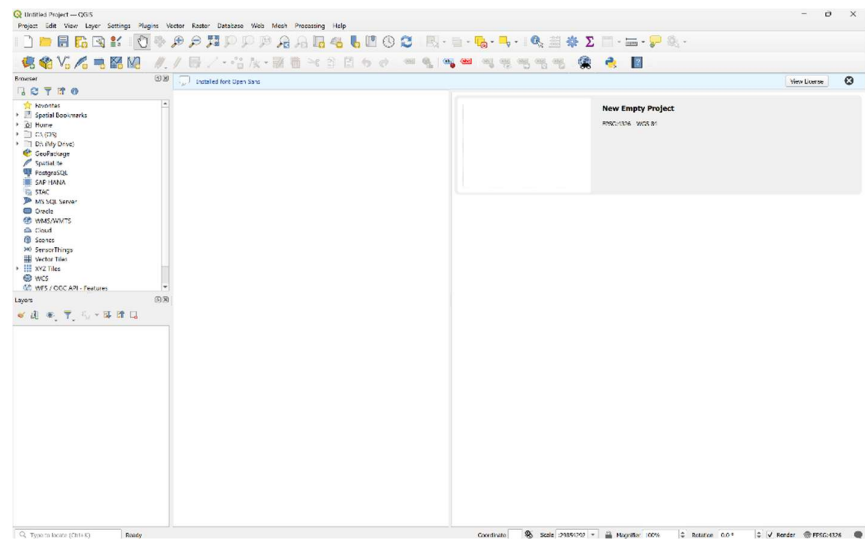


Figure 3.4: Initial QGIS Interface after Installation

3.5 QGIS User Interface

The QGIS user interface provides an integrated environment for viewing, editing, analyzing, and managing geospatial data. The interface consists of several panels, toolbars, and windows that facilitate efficient GIS operations. Understanding the different components of the QGIS interface is essential for effective data processing and analysis.

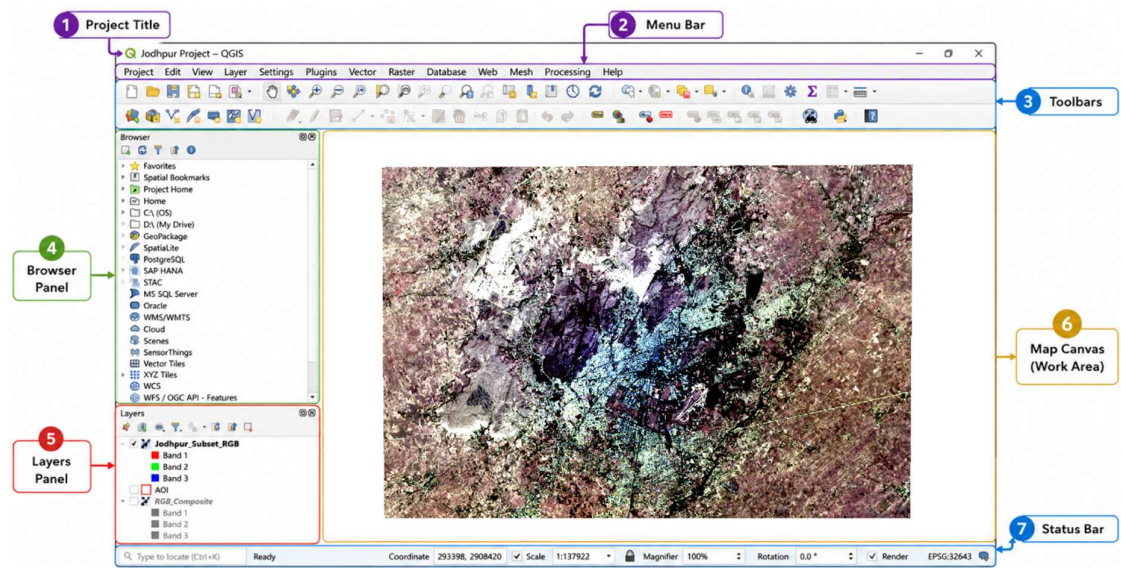


Figure 3.5: Main Components of the QGIS User Interface

3.5.1 Project Title

The Project Title is displayed on the title bar at the top of the QGIS window. It indicates the name of the currently opened QGIS project and helps users identify the active workspace.

The Project Title also assists in project organization by enabling users to distinguish between multiple QGIS projects and maintain proper workflow management during geospatial data processing.

3.5.2 Menu Bar

The Menu Bar is located at the top of the QGIS window and provides access to all major functions available within the software. Common menus include Project, Edit, View, Layer, Settings, Plugins, Vector, Raster, Processing, and Help.

The Menu Bar serves as the primary access point for project management, data processing, visualization, and analysis operations.

3.5.3 Toolbars

Toolbars provide quick access to frequently used tools and commands. QGIS contains multiple toolbars dedicated to navigation, editing, digitization, measurement, data management, and processing operations.

The use of toolbars improves workflow efficiency by reducing the need to repeatedly access menu commands.

3.5.4 Browser Panel

The Browser Panel enables users to navigate through directories and access geospatial datasets directly from the file system. It facilitates efficient management of raster and vector data without requiring manual import operations.

The Browser Panel also provides access to databases, web services, and remote data sources.

3.5.5 Layers Panel

The Layers Panel displays all spatial layers loaded into the current project. It allows users to control layer visibility, ordering, symbology, and properties.

Raster layers, vector layers, and derived datasets can be organized and managed through the Layers Panel.

3.5.6 Map Canvas

The Map Canvas is the central display area where spatial data are visualized and analyzed. Users can zoom, pan, select features, and perform various editing and analysis operations within this workspace.

The Map Canvas provides a graphical representation of all layers currently loaded into the project.

3.5.7 Status Bar

The Status Bar is located at the bottom of the QGIS interface and displays information such as map scale, coordinate position, projection information, rendering status, and task progress.

It provides useful feedback regarding ongoing operations and project settings.

3.6 Project Creation and Data Organization

Proper project organization is an important aspect of GIS-based studies. A systematic project structure ensures efficient storage, retrieval, and management of satellite imagery, processing outputs, and report documents.

3.6.1 Creation of a New QGIS Project

A new QGIS project was created to manage all spatial datasets and processing activities associated with the study. The project file served as the central workspace for loading satellite imagery, generating RGB composites, creating study area subsets, and preparing datasets for subsequent analyses.

The project was saved using the filename:

Jodhpur_Project.qgz

Saving the project file preserved layer configurations, symbology settings, coordinate systems, and workspace layouts.

3.6.2 Project Folder Structure

A structured directory system was created to organize the various datasets, outputs, and project files generated during the study. The folder hierarchy was designed to separate raw satellite imagery from processed outputs and analysis results.

The project folder structure adopted for the study is shown below:

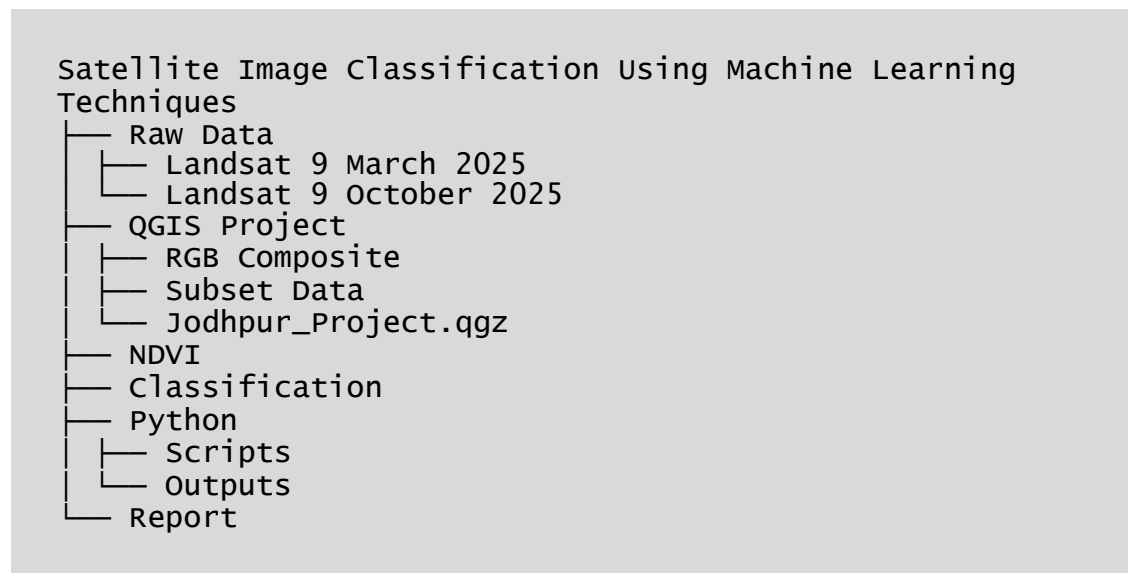


Figure 3.6: Project Folder Structure Adopted for the Study

The folder structure facilitated systematic storage of raw data, processed imagery, project files, analysis outputs, and Python implementations. Such organization minimizes data redundancy and simplifies project management.

3.6.3 Saving and Managing Project Files

During project execution, all datasets and outputs were saved in their respective directories to maintain consistency and prevent accidental data loss. Separate folders were maintained for RGB composites, study area subsets, NDVI outputs, classification results, and Python scripts.

Regular project backups were also maintained to ensure data security and reproducibility of results.

The use of an organized directory structure improved workflow efficiency and facilitated seamless integration between QGIS-based processing and Python-based analysis tasks.

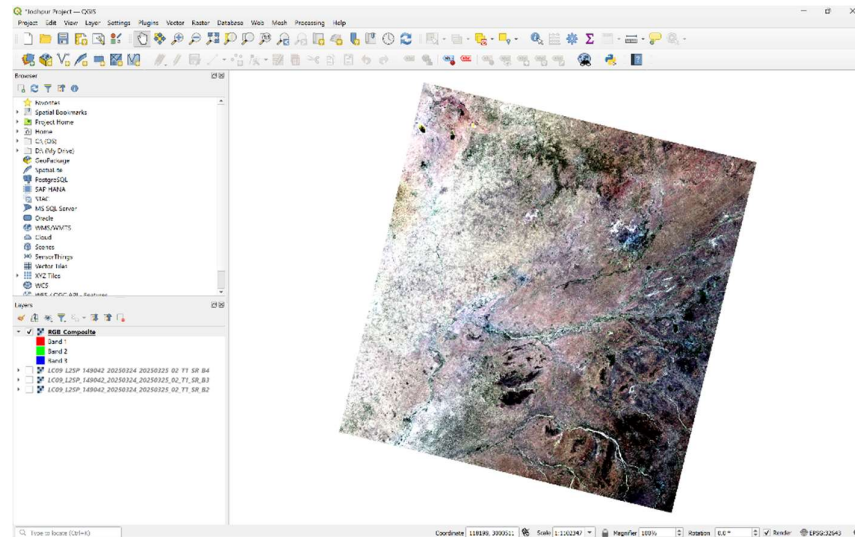


Figure 3.7: Saved QGIS Project Containing Landsat 9 Data and Project Layers

3.7 Importing Landsat 9 Data into QGIS

After downloading and extracting the Landsat 9 dataset, the individual spectral bands were imported into QGIS for visualization and processing. Loading the raster datasets into QGIS enabled further operations such as RGB composite generation, study area subsetting, vegetation analysis, and image classification.

3.7.1 Loading Raster Data

Raster data represent geographic information in the form of pixels arranged in rows and columns. Landsat 9 imagery is distributed as individual raster files corresponding to different spectral bands.

The raster layers were imported into QGIS using the "Add Raster Layer" tool available in the Layer menu. Each band was loaded separately into the project workspace.

The procedure followed was:

1. Open QGIS.
2. Navigate to Layer → Add Layer → Add Raster Layer.
3. Browse to the extracted Landsat 9 data folder.
4. Select the required spectral bands.
5. Load the raster files into the project.

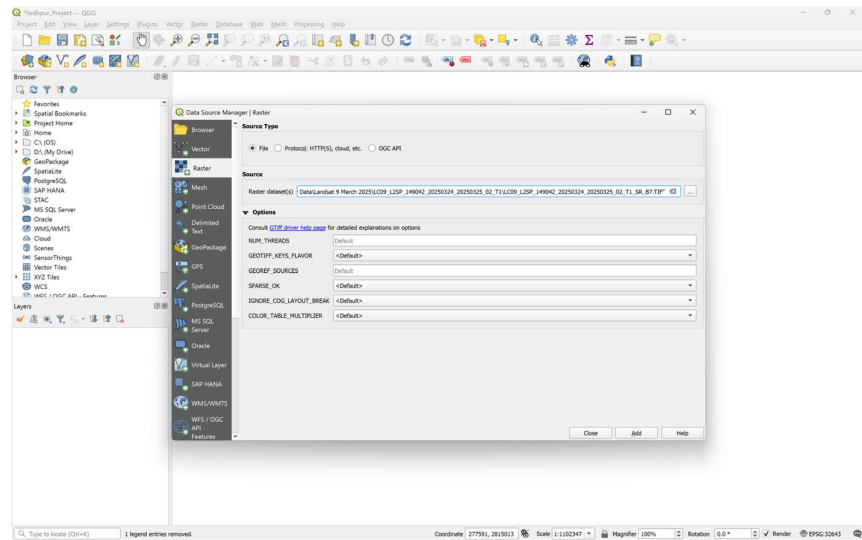


Figure 3.8: Importing Landsat 9 Raster Data into QGIS

3.7.2 Importing Landsat Bands

The downloaded Landsat 9 dataset contained multiple spectral bands covering different regions of the electromagnetic spectrum. For RGB composite generation, the following bands were selected:

Table 3.1: Landsat 9 Bands Used for RGB Composite Generation

Band	Description	Spatial Resolution
Band 2	Blue	30 m
Band 3	Green	30 m
Band 4	Red	30 m

Additional bands such as Near Infrared (Band 5) and Shortwave Infrared (Bands 6 and 7) were retained for later stages of NDVI computation and image classification.

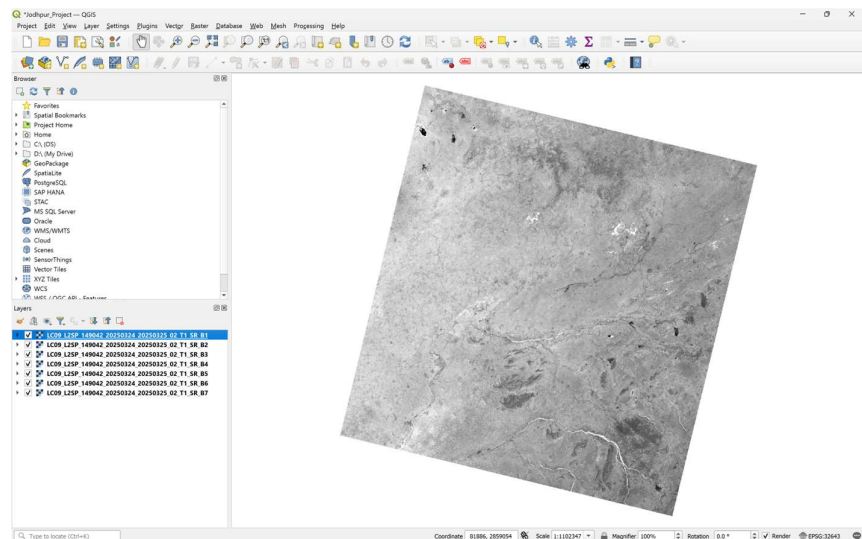


Figure 3.9: Landsat 9 Spectral Bands Loaded in QGIS

3.7.3 Verification of Raster Layers

After loading the raster datasets, the properties of each raster layer were verified to ensure data integrity and compatibility. Important parameters examined included:

- Spatial resolution
- Raster dimensions
- Coordinate Reference System (CRS)
- Pixel values
- Metadata information

Successful verification confirmed that the downloaded data were suitable for subsequent image processing operations.

3.7.4 Coordinate Reference System (CRS)

A Coordinate Reference System (CRS) defines how geographic locations are represented within a spatial dataset. Proper CRS management is essential to ensure accurate spatial analysis and alignment of multiple datasets.

The Landsat 9 imagery used in this study was referenced in:

EPSG:32643 (WGS 84 / UTM Zone 43N)

This projection system is suitable for the geographical location of Jodhpur district and enables accurate distance and area measurements.

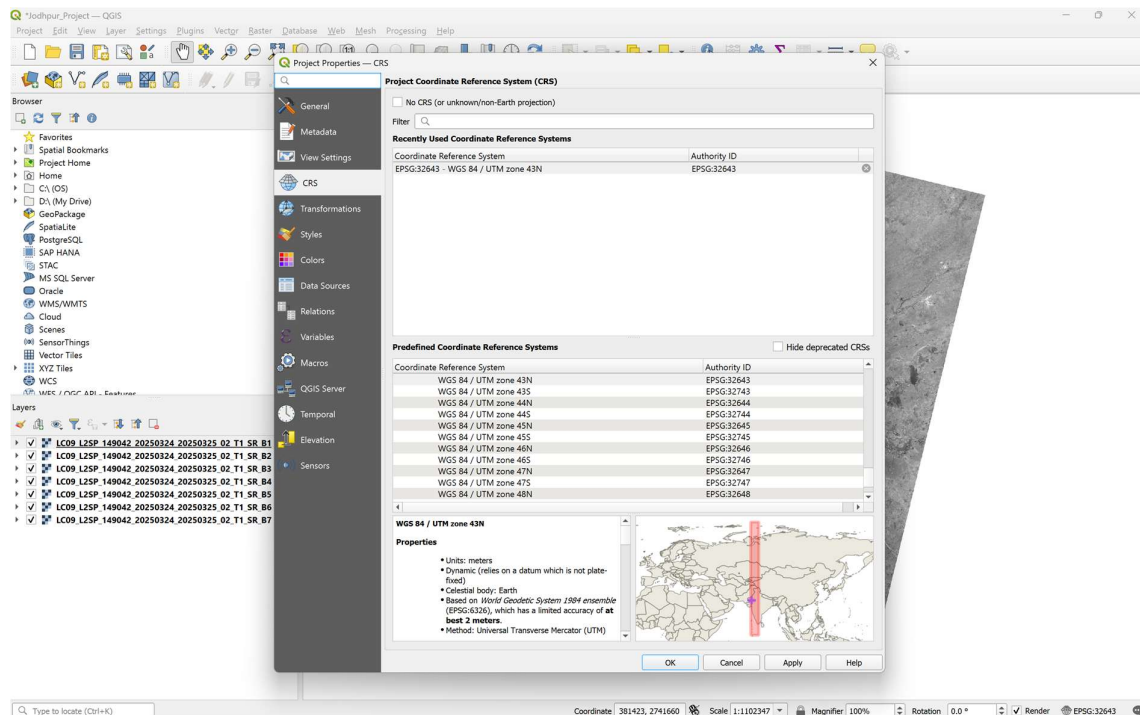


Figure 3.10: Coordinate Reference System Information of Landsat 9 Data

3.8 RGB Composite Generation

3.8.1 Introduction to RGB Composite

An RGB composite is a color image generated by assigning different spectral bands of satellite imagery to the red, green, and blue display channels. RGB composites facilitate visual interpretation of land-cover features and enable identification of vegetation, water bodies, urban areas, and barren land.

RGB composites are among the most commonly used image visualization techniques in remote sensing studies.

3.8.2 Selection of Landsat Bands

For the present study, a Natural Color Composite was generated using the following band combination:

Table 3.2: Band Combination Used for RGB Composite Generation

Display Channel	Landsat 9 Band
Red	Band 4
Green	Band 3
Blue	Band 2

This combination produces an image that closely resembles the appearance of features as observed by the human eye.

3.8.3 Creation of RGB Composite in QGIS

The RGB composite was created using the Build Virtual Raster (VRT) tool available in QGIS.

The procedure adopted was:

1. Open Raster → Miscellaneous → Build Virtual Raster.
2. Select Band 4, Band 3, and Band 2.
3. Enable the option "Place each input file into a separate band."
4. Generate the virtual raster.
5. Open Layer Properties → Symbology.
6. Assign:
 - Red Channel → Band 1 (Band 4)
 - Green Channel → Band 2 (Band 3)
 - Blue Channel → Band 3 (Band 2)
7. Apply the settings to generate the RGB composite.

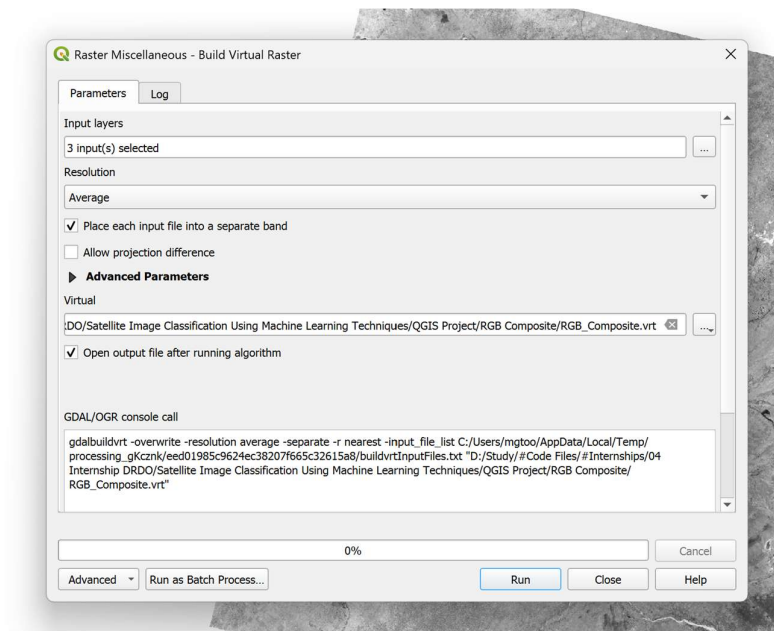


Figure 3.11: RGB Composite Generation Using Build Virtual Raster Tool

3.8.4 Visualization and Interpretation of RGB Composite

The generated RGB composite enabled visual identification of major land-cover classes within the study area.

The following features were observed:

- Built-up areas appeared as light-colored urban regions.
- Vegetation appeared as green agricultural patches.
- Water bodies appeared as dark-colored features.
- Mining and barren lands appeared as bright exposed surfaces.

The RGB composite served as the basis for selecting the Area of Interest (AOI) and extracting the study area subset used in subsequent analyses.

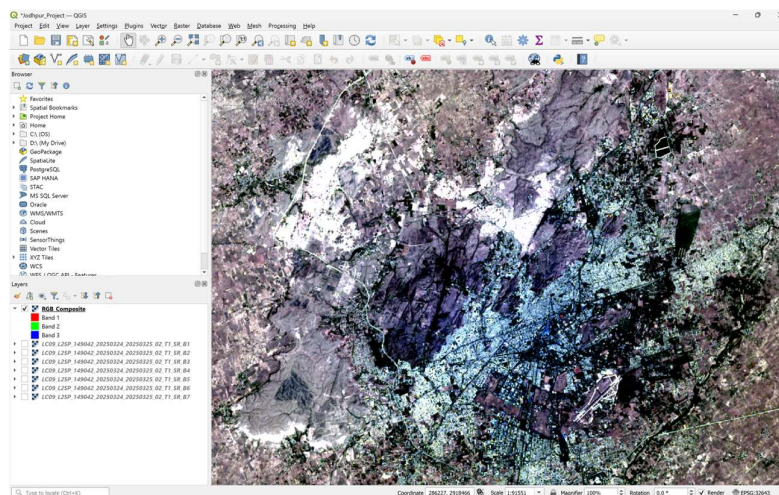


Figure 3.12: RGB Composite Image of the Jodhpur Region

3.9 Study Area Subsetting in QGIS

The Landsat 9 scene downloaded for the study covered a considerably larger geographical area than required for the present analysis. Therefore, a subset of the image was extracted to define the final study area. Subsetting reduces data volume, improves processing efficiency, and enables focused analysis of the selected region.

In this study, the subset was created using an Area of Interest (AOI) polygon and the raster clipping tools available in QGIS.

3.9.1 Creation of Area of Interest (AOI)

An Area of Interest (AOI) was created using a polygon layer in QGIS. The AOI represented the geographical boundary of the selected study region within the larger Landsat 9 scene.

The AOI was manually delineated over the RGB composite image to ensure inclusion of all major land-cover categories required for subsequent analysis.

The following land-cover classes were considered during AOI selection:

- Built-up area
- Vegetation
- Water bodies
- Mining regions
- Barren land

The inclusion of multiple land-cover categories ensures that the extracted subset is suitable for image classification and vegetation analysis.

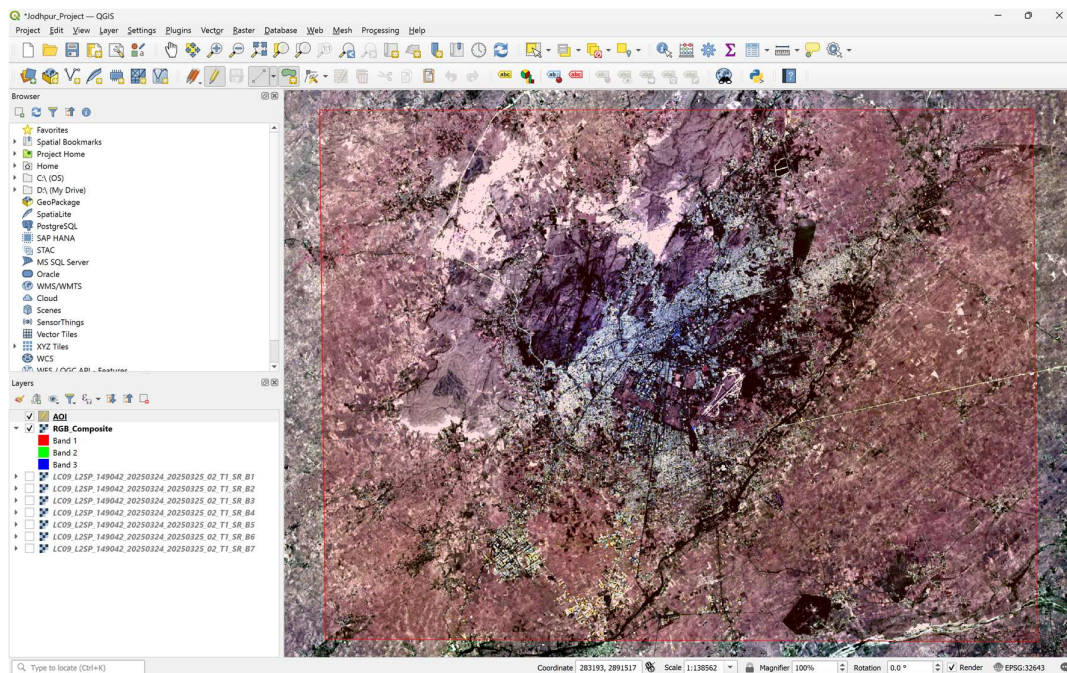


Figure 3.13: Creation of Area of Interest (AOI) on RGB Composite Image

3.9.2 Delineation of Study Area Boundary

The study area boundary was carefully defined to encompass representative samples of the major land-cover classes present in the Jodhpur region. The selected boundary included urban settlements, agricultural fields, water bodies, mining regions, and surrounding barren terrain.

The RGB composite image facilitated visual interpretation of these features and assisted in accurate AOI delineation.

The selected boundary was stored as a vector layer for use in raster extraction operations.

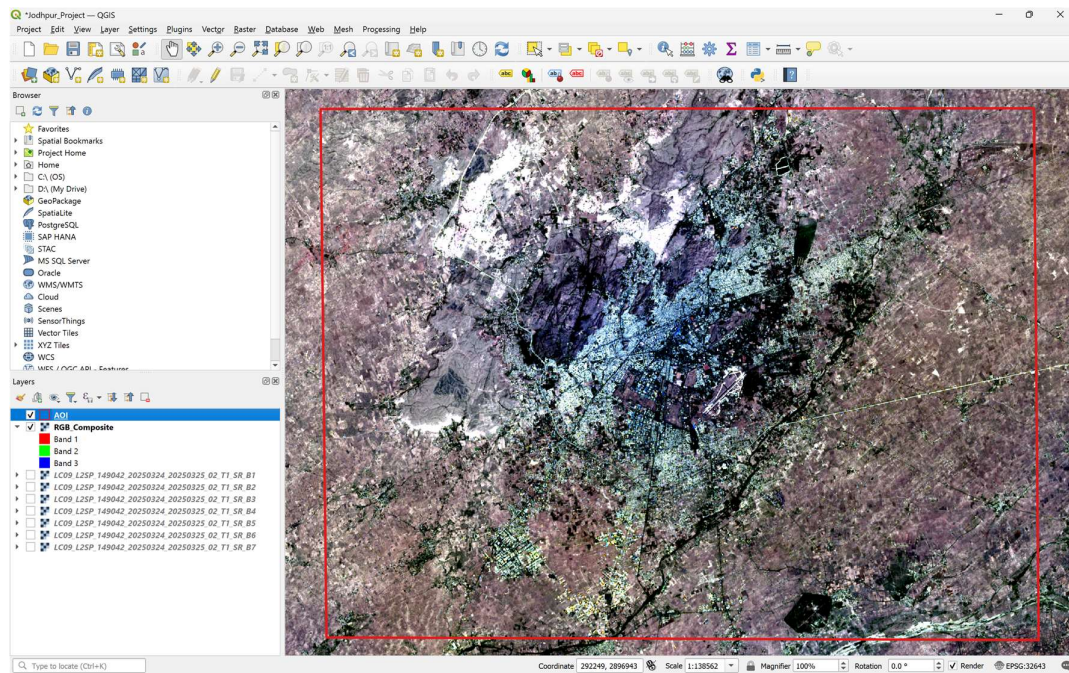


Figure 3.14: Delineated Study Area Boundary Covering Multiple Land-Cover Classes

3.9.3 Raster Clipping Using AOI Mask Layer

After defining the AOI, the RGB composite image was clipped using the “Clip Raster by Mask Layer” tool available in QGIS.

The clipping process involved:

1. Selecting the RGB composite as the input raster.
2. Selecting the AOI polygon as the mask layer.
3. Enabling clipping based on the AOI extent.
4. Generating the output raster containing only the selected study area.

This operation removed unwanted regions outside the study boundary and produced a raster dataset focused exclusively on the area of interest.

The clipping process significantly reduced the dataset size and improved processing efficiency for subsequent analyses.

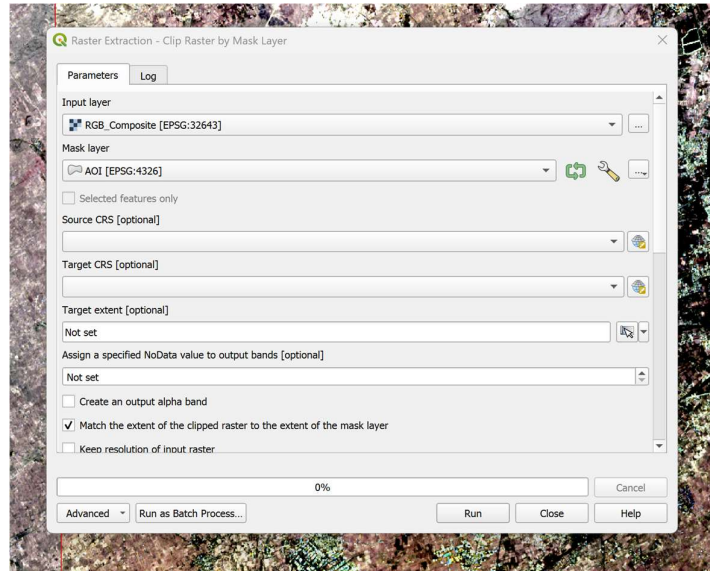


Figure 3.15: Clip Raster by Mask Layer Tool Used for Study Area Extraction

3.9.4 Generation of Final Study Area Subset

The clipping operation produced the final study area subset, which served as the primary dataset for all remaining stages of the project.

The extracted subset contained representative samples of:

- Built-up land
- Vegetation
- Water bodies
- Mining areas
- Barren terrain

The subset image formed the basis for NDVI computation, image classification, and machine learning-based analysis performed in later chapters.

The final subset was stored as a separate raster file and organized within the project directory for future processing.

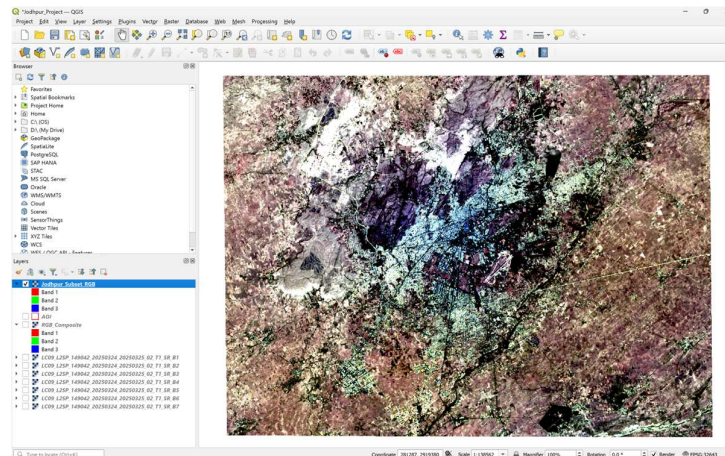


Figure 3.16: Final Study Area Subset Extracted from Landsat 9 Imagery

3.10 Summary

This chapter introduced the concepts of Geographic Information Systems (GIS) and described the role of GIS in remote sensing applications. The chapter presented the components, functions, and major application areas of GIS and discussed the significance of QGIS as an open-source geospatial software platform.

The installation and configuration of QGIS were explained, followed by an overview of the QGIS user interface and project organization strategy adopted for the study. The procedures for importing Landsat 9 raster data, managing coordinate reference systems, and generating RGB composite images were also described.

Furthermore, the chapter detailed the creation of an Area of Interest (AOI), study area delineation, raster clipping, and generation of the final study area subset. The processed subset generated in this chapter serves as the primary dataset for the NDVI analysis and image classification tasks presented in the subsequent chapters.

3.11 References

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